

PHK Early Warning Dashboard

Technical Notes

July 19, 2026

Contents

1	Methodology for Provincial PHK Analysis	3
1.1	Objective	3
1.2	Econometric Approach	3
1.3	Rationale for Model Selection	4
1.4	Variables	5
2	Results	7
2.1	Macroeconomic Pressure and Leading Indicator Analysis	7
2.2	Poisson Regression: Single-Lag Specifications	8

1 Methodology for Provincial PHK Analysis

1.1 Objective

The objective of the analysis is to examine how macroeconomic pressures are associated with the monthly number of layoffs (PHK) across provinces, while accounting for differences in provincial economic and labor-market structure. The framework combines structural characteristics of provincial economies and labor markets with high-frequency macroeconomic indicators to capture both underlying structural vulnerability and short-term economic pressures. Lagged macroeconomic indicators are used to assess whether changes in macroeconomic conditions precede subsequent changes in PHK.

1.2 Econometric Approach

- The dependent variable is the monthly number of reported layoffs (PHK) by province, which is a non-negative count variable with a substantial number of zero observations.
- A Poisson regression with province fixed effects and calendar-month fixed effects is employed. The model specification is as follows:

$$E(PHK_{it} | M_{i,t-k}, S_{it}) = \exp(\beta M_{i,t-k} + \theta' S_{it} + \alpha_i + \delta_{m(t)}), \quad (1)$$

where:

- PHK_{it} denotes the number of reported layoffs in province i during month t ;
 - $M_{i,t-k}$ denotes a macroeconomic pressure indicator observed k months prior to month t . Depending on the indicator, macroeconomic pressures are measured at the global, national, or provincial level;
 - S_{it} denotes provincial economic and labor-market structural characteristics included as control variables;
 - α_i denotes province fixed effects, which control for time-invariant differences across provinces;
 - $\delta_{m(t)}$ denotes calendar-month fixed effects (January–December), which control for recurring seasonal patterns in PHK; and
 - $k \in \{1, 3, 6, 12\}$ denotes the lag horizon, measured in months.
- Macroeconomic pressure indicators are measured at the global, national, or provincial level depending on data availability and the economic mechanism considered. The indicators enter the regressions at monthly frequency and are evaluated at lag horizons of 1, 3, 6, and 12 months.
 - Provincial structural characteristics capture underlying differences in economic and labor-market conditions across provinces. Economic structure variables include indicators such as sectoral composition of GRDP, investment and trade exposure, GRDP per capita, and poverty. Labor-market structure variables include indicators such as unemployment, underemployment, employment composition, formal employment, wages, and labor-force participation.

- Provincial structural variables are measured annually and assigned to the corresponding monthly observations within each year. These variables are included primarily as controls for differences in underlying provincial economic structure and labor-market vulnerability rather than interpreted as causal determinants of PHK.
- Two complementary specifications are estimated. First, each macroeconomic pressure indicator is entered separately, together with the common set of provincial structural controls, to assess its individual association with subsequent PHK. Second, all macroeconomic pressure indicators are entered jointly in single-lag specifications to assess their conditional associations while holding other observed macroeconomic pressures constant. Lag horizons of 1, 3, 6, and 12 months are considered separately.
- Robust standard errors are clustered at the provincial level to account for within-province serial correlation.

1.3 Rationale for Model Selection

- **Poisson regression is appropriate for the count nature of the dependent variable.**

The dependent variable is the monthly number of reported layoffs (PHK) by province and is therefore a non-negative count variable that includes zero observations. A Poisson specification directly models the conditional mean of the count outcome and ensures non-negative fitted values. Importantly, consistency of the Poisson estimator does not require the dependent variable to follow an exact Poisson distribution, provided that the conditional mean is correctly specified. Robust inference is used to accommodate potential violations of the Poisson variance assumption ([Hausman et al., 1984](#); [Cameron and Trivedi, 2013](#)).

- **Province fixed effects account for persistent unobserved heterogeneity across provincial labour markets.**

Provinces differ systematically in industrial composition, economic structure, labour-market institutions, geography, and reporting practices. Some of these persistent characteristics may be correlated with the observed explanatory variables and with the incidence of reported PHK. Province fixed effects control for time-invariant provincial characteristics, allowing identification to rely on changes in explanatory variables within provinces over time ([Hausman et al., 1984](#); [Wooldridge, 2010](#)).

- **The panel structure allows macroeconomic pressures to be examined while accounting for differences in provincial exposure and vulnerability.**

The analysis combines monthly PHK observations with macroeconomic pressure indicators and provincial economic and labour-market characteristics. This framework is suitable for assessing whether changes in macroeconomic conditions precede subsequent changes in PHK while accounting for systematic differences in provincial structure. Lagged macroeconomic indicators at 1, 3, 6, and 12 months are examined to capture potentially different transmission horizons.

- **Calendar-month fixed effects account for recurring seasonal patterns without absorbing aggregate macroeconomic variation.**

Calendar-month fixed effects (January–December) are included to control for recurring seasonal patterns in reported PHK that may systematically differ across months of the year. Unlike a full set of month-year fixed effects, calendar-month fixed effects preserve the time-series variation in macroeconomic indicators that is required to estimate their association with PHK.

- **Clustered standard errors account for within-province dependence over time.**

Standard errors are clustered at the province level, allowing for heteroskedasticity and arbitrary correlation of regression disturbances within provinces over time. This provides inference that is robust to serial dependence within provincial panels.

1.4 Variables

- Dependent Variable
 - Monthly Provincial PHK Flow (number of reported layoffs).
- Provincial Economic Structure (selected using PCA)
 - Share of Government Expenditure in GRDP (%)
 - Share of Investment in GRDP (%)
 - Share of Exports in GRDP (%)
 - Share of Imports in GRDP (%)
 - Share of Manufacturing in GRDP (%)
 - Share of Agriculture in GRDP (%)
 - Share of Mining in GRDP (%)
 - GRDP per Capita (log)
 - Poverty Rate (%)
- Provincial Labor Market Structure (selected using PCA)
 - Unemployment Rate (%)
 - Underemployment Share (%)
 - Labor Force Participation Rate (%)
 - Formal Labor Share (%)
 - Manufacturing Labor Share (%)
 - Agriculture Labor Share (%)
 - Full-Time Worker Share (%)
 - Minimum Wage (% yoy)
 - Average Wage (log)

- Macroeconomic Pressure
 - Global
 - * Brent Crude Oil Price (log)
 - National
 - * BI Rate (%)
 - * IDR/USD Exchange Rate (log)
 - * Manufacturing PMI
 - * IHPB - Wholesale Price Index
 - Provincial
 - * IHK - Consumer Price Index
 - * Non-Performing Loans (NPL)/Credit Ratio
 - * Export Value (log)
 - * Import Value (log)

2 Results

2.1 Macroeconomic Pressure and Leading Indicator Analysis

Table 1: Direction of Significant Associations between Lagged Macroeconomic Indicators and PHK

Variable	Lag 1	Lag 3	Lag 6	Lag 12
log_price_brent	+			
bi_rate	+	+	+	+
log_fx_idr_usd		-		
pmi_manuf				
ihpb_yoy		-		-
cpi_yoy	-	-	-	-
npl_ratio				
log_export	-	-		
log_import	-			

Notes: Each cell summarizes the estimated association from a separate Poisson regression of monthly provincial PHK on the indicated lagged macroeconomic pressure variable, controlling for provincial economic and labor-market structure, province fixed effects, and calendar-month fixed effects. '+' ('-') denotes a positive (negative) association statistically significant at the 10% level or better. Blank cells denote estimates that are not statistically significant at the 10% level.

Table 2: Incidence Rate Ratios (IRRs) from Poisson Regressions

Variable	Lag 1	Lag 3	Lag 6	Lag 12
log_price_brent	21.08***	2.16	0.59	3.21
bi_rate	2.83***	2.89***	1.89***	1.61***
log_fx_idr_usd	0.71	0.01***	7.49	27.83
pmi_manuf	0.95	0.99	1.01	1.03
ihpb_yoy	0.85	0.66***	1.09	0.51**
cpi_yoy	0.88**	0.85**	0.81***	0.59***
npl_ratio	0.84	0.85	0.73	0.61
log_export	0.67*	0.36**	1.16	1.02
log_import	0.83*	0.93	0.93	1.12

Notes: Each cell reports the incidence rate ratio (IRR) from a separate Poisson regression of monthly provincial PHK on the indicated lagged macroeconomic pressure variable, controlling for provincial economic and labor-market structure, province fixed effects, and calendar-month fixed effects. IRRs are obtained by exponentiating the estimated Poisson coefficients. An IRR above (below) one indicates a positive (negative) association with expected PHK. The magnitude of the IRR should be interpreted according to the unit and transformation of each explanatory variable. Standard errors are clustered at the province level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

2.2 Poisson Regression: Single-Lag Specifications

Table 3: Poisson Regression Estimates of PHK on Macroeconomic Pressures

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Macroeconomic Pressure (with lag)				
L_log_price_brent	1.310 (1.846)	1.893 (1.161)	-0.001 (1.759)	2.437 (3.170)
L_bi_rate	1.033 (0.571)	0.880* (0.354)	1.621** (0.555)	4.002*** (0.798)
L_log_fx_idr_usd	-2.430 (6.533)	-1.692 (2.903)	-5.278 (6.020)	-16.039** (5.625)
L_pmi_manuf	-0.047 (0.056)	-0.024 (0.073)	-0.114 (0.089)	-0.105 (0.204)
L_ihpb_yoy	0.133 (0.257)	-0.633*** (0.166)	0.151 (0.279)	-1.151* (0.569)
L_cpi_yoy	0.014 (0.146)	0.280* (0.135)	-0.434** (0.151)	-0.168 (0.176)
L_npl_ratio	0.066 (0.345)	0.447 (0.326)	0.412 (0.364)	-0.491 (0.392)
L_log_export	-0.247 (0.246)	-0.667 (0.479)	0.377 (0.630)	0.497 (0.470)
L_log_import	-0.225 (0.117)	-0.101 (0.107)	-0.011 (0.138)	0.159 (0.085)
Provincial Economic Structure				
gov_pdrb_share	0.050 (0.285)	-0.157 (0.285)	-0.050 (0.230)	0.122 (0.450)
invest_pdrb_share	-0.137 (0.162)	0.060 (0.167)	0.066 (0.186)	0.360* (0.153)
export_pdrb_share	-0.105 (0.308)	0.237 (0.159)	0.108 (0.164)	-0.379 (0.564)
import_pdrb_share	0.456 (0.477)	0.771* (0.360)	0.707 (0.414)	1.055* (0.417)
manuf_pdrb_share	-0.109 (0.265)	-0.104 (0.220)	0.167 (0.193)	0.769* (0.327)
agri_pdrb_share	0.529 (0.858)	0.114 (0.726)	0.707 (0.880)	2.962* (1.281)
mining_pdrb_share	0.075 (0.465)	-0.025 (0.243)	0.095 (0.268)	1.151 (0.659)
log_pdrb_cap	9.759 (10.545)	-3.317 (8.257)	-7.238 (7.027)	-34.843* (13.705)
pov_rate	-0.636 (0.427)	-0.302 (0.400)	-0.379 (0.409)	-0.295 (0.293)
Provincial Labor Market Structure				
unemployment_rate	0.635 (0.791)	0.767 (0.775)	0.382 (0.601)	1.663 (1.078)
underemployment_share	0.231 (0.200)	0.039 (0.155)	0.112 (0.140)	0.773** (0.286)
labor_force_rate	-0.031 (0.192)	-0.072 (0.203)	-0.138 (0.216)	0.229 (0.183)
formal_lab_share	0.020 (0.133)	-0.065 (0.127)	-0.050 (0.127)	0.381*** (0.113)
manuf_emp_share	0.269 (0.390)	-0.021 (0.313)	-0.136 (0.430)	-1.178*** (0.345)
agri_emp_share	0.120 (0.179)	0.127 (0.156)	0.101 (0.203)	-0.551*** (0.151)
full_time_share	-0.061 (0.128)	-0.070 (0.119)	-0.041 (0.115)	-0.012 (0.106)
min_wage_yoy	0.106 (0.086)	0.037 (0.091)	-0.065 (0.106)	-0.216 (0.129)
log_avg_wage	4.766 (4.996)	2.273 (4.346)	-0.555 (4.733)	-2.836 (4.130)
Observations	1003	948	865	696
Monthly PHK Mean	223	224	232	235
Pseudo R ²	0.600	0.616	0.631	0.695
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: The dependent variable is the monthly number of PHK by province. Columns (1)–(4) report separate Poisson specifications in which all macroeconomic pressure indicators enter at lags of 1, 3, 6, and 12 months, respectively. Provincial economic and labor-market structural controls enter contemporaneously. All specifications include province and calendar-month fixed effects. Standard errors clustered at the province level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

Table 4: Incidence Rate Ratios (IRR) from Poisson Regression Estimates of PHK on Macroeconomic Pressures

	(1) Lag 1	(2) Lag 3	(3) Lag 6	(4) Lag 12
Macroeconomic Pressure (with lag)				
L_log_price_brent	3.70 (6.84)	6.64 (7.71)	1.00 (1.76)	11.44 (36.27)
L_bi_rate	2.81 (1.61)	2.41* (0.85)	5.06** (2.81)	54.70*** (43.65)
L_log_fx_idr_usd	0.09 (0.57)	0.18 (0.53)	0.01 (0.03)	0.00** (0.00)
L_pmi_manuf	0.95 (0.05)	0.98 (0.07)	0.89 (0.08)	0.90 (0.18)
L_ihpb_yoy	1.14 (0.29)	0.53*** (0.09)	1.16 (0.32)	0.32* (0.18)
L_cpi_yoy	1.01 (0.15)	1.32* (0.18)	0.65** (0.10)	0.85 (0.15)
L_npl_ratio	1.07 (0.37)	1.56 (0.51)	1.51 (0.55)	0.61 (0.24)
L_log_export	0.78 (0.19)	0.51 (0.25)	1.46 (0.92)	1.64 (0.77)
L_log_import	0.80 (0.09)	0.90 (0.10)	0.99 (0.14)	1.17 (0.10)
Provincial Economic Structure				
gov_pdrb_share	1.05 (0.30)	0.85 (0.24)	0.95 (0.22)	1.13 (0.51)
invest_pdrb_share	0.87 (0.14)	1.06 (0.18)	1.07 (0.20)	1.43* (0.22)
export_pdrb_share	0.90 (0.28)	1.27 (0.20)	1.11 (0.18)	0.68 (0.39)
import_pdrb_share	1.58 (0.75)	2.16* (0.78)	2.03 (0.84)	2.87* (1.20)
manuf_pdrb_share	0.90 (0.24)	0.90 (0.20)	1.18 (0.23)	2.16* (0.70)
agri_pdrb_share	1.70 (1.46)	1.12 (0.81)	2.03 (1.78)	19.33* (24.77)
mining_pdrb_share	1.08 (0.50)	0.98 (0.24)	1.10 (0.29)	3.16 (2.09)
log_pdrb_cap	17311.68 (182559.68)	0.04 (0.30)	0.00 (0.01)	0.00* (0.00)
pov_rate	0.53 (0.23)	0.74 (0.30)	0.68 (0.28)	0.74 (0.22)
Provincial Labor Market Structure				
unemployment_rate	1.89 (1.49)	2.15 (1.67)	1.46 (0.88)	5.27 (5.68)
underemployment_share	1.26 (0.25)	1.04 (0.16)	1.12 (0.16)	2.17** (0.62)
labor_force_rate	0.97 (0.19)	0.93 (0.19)	0.87 (0.19)	1.26 (0.23)
formal_lab_share	1.02 (0.14)	0.94 (0.12)	0.95 (0.12)	1.46*** (0.17)
manuf_emp_share	1.31 (0.51)	0.98 (0.31)	0.87 (0.38)	0.31*** (0.11)
agri_emp_share	1.13 (0.20)	1.14 (0.18)	1.11 (0.22)	0.58*** (0.09)
full_time_share	0.94 (0.12)	0.93 (0.11)	0.96 (0.11)	0.99 (0.11)
min_wage_yoy	1.11 (0.10)	1.04 (0.09)	0.94 (0.10)	0.81 (0.10)
log_avg_wage	117.43 (586.73)	9.71 (42.20)	0.57 (2.72)	0.06 (0.24)
Observations	1003	948	865	696
Monthly PHK Mean	223	224	232	235
Pseudo R ²	0.600	0.616	0.631	0.695
Region FE	Province	Province	Province	Province
Time FE	Month	Month	Month	Month

Notes: Entries report incidence rate ratios (IRRs), obtained by exponentiating the corresponding Poisson coefficients. Columns (1)–(4) jointly include all macroeconomic pressure indicators at lags of 1, 3, 6, and 12 months, respectively, together with contemporaneous provincial economic and labor-market structural controls, province fixed effects, and calendar-month fixed effects. An IRR above (below) one indicates a positive (negative) conditional association with expected PHK. The magnitude of each IRR should be interpreted according to the unit and transformation of the corresponding explanatory variable. Standard errors clustered at the province level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

References

- Cameron, A. C. and Trivedi, P. K. (2013). Regression Analysis of Count Data. Cambridge University Press, 2 edition.
- Hausman, J. A., Hall, B. H., and Griliches, Z. (1984). Econometric models for count data with an application to the patents–r&d relationship. Econometrica, 52(4):909–938.
- Wooldridge, J. M. (2010). Econometric Analysis of Cross Section and Panel Data. MIT Press, 2 edition.